MSX and SPECTRAVIDEO

COMPUTER FORUM

Volume 1 No. 11

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MSX and SPECTRAVIDEO COMPUTER FORUM

THE MAGAZINE FOR MSX AND SPEC… …MPUTERS

As the name suggests, CO… …rum for exchange of information and … computer.

Reg…

… *manual leaves*

… *it.*

… *r,*

PLUS –

Please regist…

I enclose Chec…

OR

Charge my Ban…

Bankcard …

Please print clearly.

Card expiry date:……

Name:……

Address:……

……

State:…… Postcode:……

Computer make and type:……

Start Subscription

from Vol …… No ……

A LETTER FROM THE EDITOR

Dear Subscriber

For most of you, your subscription expires on receipt of the next issue of Computer Forum. I thought I would take this opportunity to have a chat about the future of MSX and Spectravideo and also this magazine.

Due to the facts of MSX dealers disappearing like your money at the races, Spectravideo having ceased production and MSX hardware no longer being imported, then the need for an independent source of information becomes absolutely essential.

There are a few changes planned for Computer Forum. Negotiations are currently under way for a substantial increase in the number of journalistic and production staff. This will inject some new blood into the system and enhance the content.

Publication of Computer Forum magazine will continue. It's future success, however, will depend on your support in the form of a subscription renewal.

THE BOTTOM LINE: We need to see the colour of your money - AGAIN!

Kind regards

Broderick J Brown
Managing Editor

MSX and SPECTRAVIDEO

COMPUTER FORUM

Magazine

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EDITORIAL

This month we put a couple of printers through their paces, look at some more software (including a few for Spectravideo), and other goodies.

The mailbox page, though, is getting thin. Remember, the operative part of our name is Forum - a place for people to meet and discuss items of common interest. I know, you're all sat their saying "I don't want to feel a gallah by telling people about my silly little problems" or "I don't have problems, I know it all".

Well, your "silly little problem" is probably shared by quite a few others. There is no such thing as an instant expert. The guy who claims to know it all is being a bit selfish because someone probably helped him overcome his silly little problems to get to his current position of satisfaction.

When you write to Forum though, please use a separate sheet for your mailbox item so that it gets attention as soon as possible. I look forward to hearing from all of you in turn - just so that I know that you are reading all of this.

Happy computing ...

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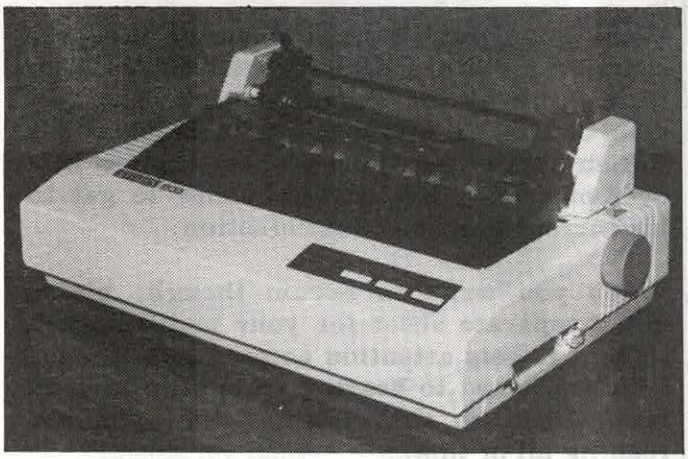
PRODUCT NEWS ...

WE COMPARE PRINTERS

We are continually being asked which printers are suitable for use with Spectravideo and MSX computers - so we decided to have a close look at a couple of popular models currently available.

The two chosen were the Panasonic KX-P1080 and the Citizen 120-D. Both are normal dot matrix printers and as such will not reproduce the graphics characters used by MSX printers. That does not mean that they do not have a graphics capability. It means that their normal character set does not contain the block graphics which can be generated from an MSX keyboard.

THE CITIZEN 120-D

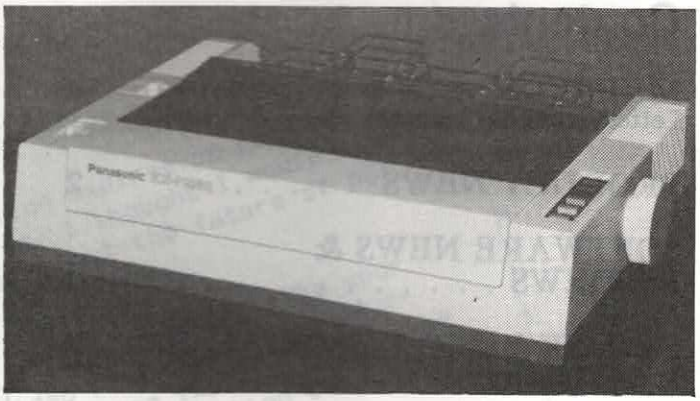


When you first see this printer, you could be put off by its compact size - 370 x 238 mm and weighing 3.7kg. It comes with a well documented manual of 184 pages but lacks a detailed index. A lift out command summary partially compensates for this. There are a great many example programs throughout.

The printer cable plugs into the right side of the printer which contains a removable cartridge housing the parallel interface and the DIP switches which are used to setup the printer for various computers. Nine print styles are selectable using a combination of presses on the Form Feed and Line Feed buttons. The price of replacement ribbons is $14.40.

Made by the Citizen Watch Co Ltd in Japan, at $449 it is well within the budget of the home enthusiast or business looking for a good quality printer.

THE PANASONIC KX-P1080



The first impression is that you have seen this printer somewhere before. In fact you probably have as it is also marketed under different brand names and its style is not very revolutionary.

This printer is styled along the same lines as much larger units and therefore looks very business-like. It is larger than the Citizen 120-D being 403 x 286 mm and weighs 6kg.

The P1080 has a well indexed 96 page A4 manual with loads of example programs to demonstrate the different print functions and styles. The printer cable plugs into the rear of the printer and the DIP switches are accessed by removing a dust cover near the print head. Three print modes - Draft, NLQ and Compressed can be selected from the front panel. A built-in ink reserve on the ribbon can be activated to extend its life and new ribbons are $17.00.

Manufactured in Japan by Matsushita (National) and priced at $499 it represents good value for both home and work.

LIKES AND DISLIKES

I liked the Citizen for the removable interface cartridge and easily adjusted DIP switches, the position of the centronics plug, its speed and compactness. The lack of an index in the manual is a pity.

I liked the rugged looks of the Panasonic, its general performance, extra life on the ribbon and the index in the manual. I did not like the plug being at the rear - it gets in the way of the paper flow like many other printers.

CONCLUSION

Without owning both printers for twelve months to average out the speed difference and ribbon costs it is difficult to make a practical choice.

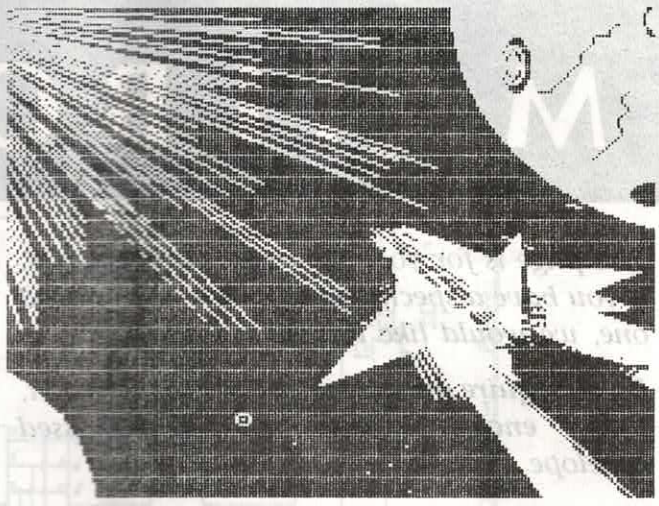
I liked both printers for different reasons. Given that both are professional quality and reasonable price, the choice would be an individual one and based on personal preference for typestyle and appearance.

COMPARISON CHART

	Citizen	Panasonic
Print speed (CPS)		
Draft mode	120	100
Near Letter Quality	24	20
Buffer capacity	4K	1K (4K opt)
Print Head	9 pin	9 pin
Max paper width	10 inch	10 inch
Ribbon life (Chr.)	2 Million	3 Million
Paper IN	rear/bottom	rear
Paper OUT	rear	rear
DIP switches	8	8
DIP access	slide tray	removable cover
Front controls	ON LINE	ON LINE
	LINE FEED	LINE FEED
	FORM FEED	FORM FEED
	FRIC/PIN	FRIC/PIN
		PRINT MODE
Indicators	Power	Power
	Paper Out	Paper/Error
	Ready	On Line
Graphics		
(Dots per inch)	60	60
	72	72
	80	80
	90	90
	120	120
	144	240
	240	

Pica Font
Elite Font
Pica Proportional
Elite Proportional
Compressed
Emphasized
Pica Doublestrike
Elite Doublestrike
Pica Expanded
Elite Expanded
Pica Italics
Elite Italics
Pica Underline
Elite Underline

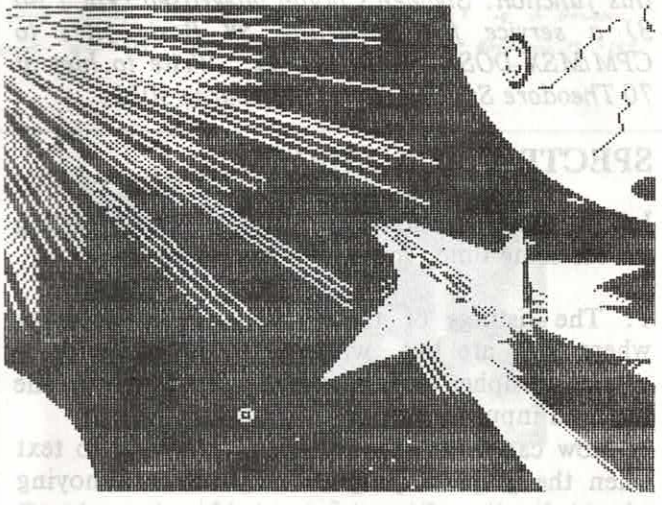
Sample print of Citizen 120-D



Graphics screen dump to Citizen 120-D

Pica Font
Elite Font
Pica proportional
Elite proportional
Compressed
Pica Doublestrike
Elite Doublestrike
Pica Expanded
Pica Expanded
Elite Expanded
Pica Italics
Elite Italics
Pica Underline
Elite Underline

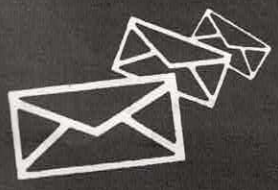
Sample print of Panasonic KX-P1080



Graphics screen dump to Panasonic KX-P1080

Review equipment from Peter Fox Computers, Fremantle.

MAILBOX



This page is for your comments and queries. If you have a specific problem or have solved one, we would like to hear about it.

If you require an answer before we publish it, please enclose a stamped self-addressed envelope.

JETSET WILLY HELP

The MSX program Jetset Willy was being sold in a gift pack with Manic Miner. The problem is that the distributors forgot to include the code book required to break the code at the start of Jetset Willy. The distributors have since ceased involvement with MSX and cannot supply the books. Can anyone help?

CP/M TO BASIC FOR USA

Can anyone direct me to a source for a program that moves ASCII stored programs (MBASIC-80) to SV Disk Basic?

John Rudzinski
Michigan, USA

We are not aware of a program which performs this function. Stephen Colgan advertised (Vol 1 No 3) a service for converting SV Basic files to CPM/MSX-DOS/MSX Disk Basic. Write to him at 70 Theodore St, St Albans, Melbourne 3021.

SPECTRAVIDEO HELP

I have four questions which have been puzzling me for some time, so perhaps you can help.

1. The listings of the ROM subroutines show where they are but, without more experience, I cannot decipher the subroutines to find out the required inputs and results.
2. How can I stop the machine returning to text when the graphics program stops? It is annoying when trying to get input for a graphics program.
3. Is there any way that sprite plane 34 (external video) can be implemented.
4. My disk drives continue for about 40 seconds after use, most annoying when other machines stop after about 3 seconds. Can this be altered?

Keep up the good work with the magazine, I look forward to it continuing for a long time yet.

Tony James
Stawell, Victoria

We are hoping to publish the information required by each ROM routine and the returned values. In the meantime, a good disassembler will enable you to investigate many for yourself. A program will always revert to text mode when you use the INPUT command. You will have to create your own routine to handle input for the graphics screen (see the Basic Tutorial in this issue). As far as we know, the external video plane can only be activated by a hardware mod.

PARALLEL TO SERIAL HELP

Enclosed is a circuit I built to convert the parallel printer output to serial format (as enquired for by Mr Creek of Glen Waverley). Made from readily available components it requires +5, +12 and -12 volts which are available from the SV expansion bus.

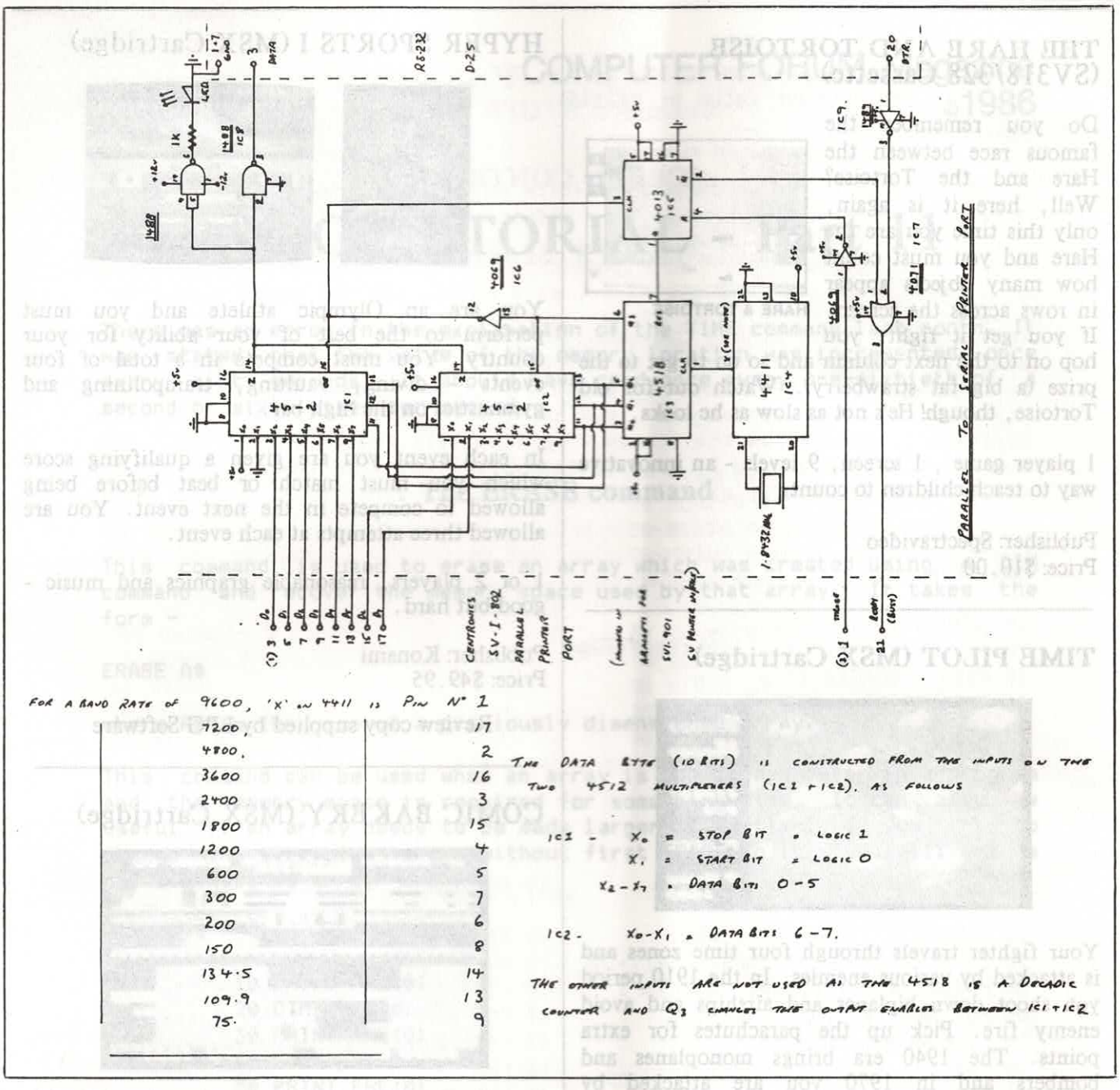
For readers already having an RS232 card, I can supply a modified Disk Basic that activates the undocumented "MDM" commands including OPEN "MDM:" AS #1, ON MDM GOSUB, MDM ON etc. with a default setting of 300 baud, 8 bit, no parity. If the RS232 connects to a printer a program listing can be obtained by using SAVE "MDM:". LPRINT can be simulated by OPEN"MDM:"AS#1 and then PRINT#1.

I also have a terminal program that allows setting the baud rate and other information of the RS232 port to allow file transfer without protocol, capture of data to the printer, screen dumping, save/load profiles, three programmable function keys and a 'BREAK' key and on-line timer.

I have the modified Basic and terminal Disk version available at $30 or $15 for modified Disk Basic alone. A tape version of both is available for $25 or $15 for the modified Basic.

R Johnson
28 Simon St, Blackburn Nth 3130

Thanks for the information. A diagram of the circuit is reproduced here.



FOR A BAUD RATE OF 9600, 'X' ON 4511 IS PIN N° 1

7200	17
4800	2
3600	16
2400	3
1800	15
1200	4
600	5
300	7
200	6
150	8
134·5	14
109·9	13
75	9

THE DATA BYTE (10 BITS) IS CONSTRUCTED FROM THE INPUTS ON THE TWO 4512 MULTIPLEXERS (IC1 + IC2). AS FOLLOWS

IC1 - X_0 = STOP BIT = LOGIC 1
X_1 = START BIT = LOGIC 0
$X_2 - X_7$ = DATA BITS 0-5

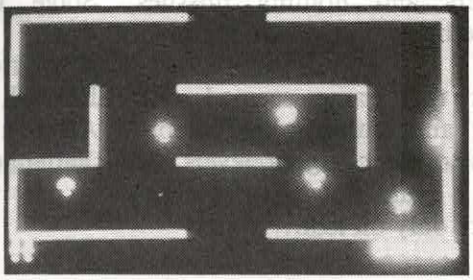
IC2 - $X_0 - X_1$ = DATA BITS 6-7.

THE OTHER INPUTS ARE NOT USED AS THE 4518 IS A DECADIC COUNTER AND Q_3 CHANGES THE OUTPUT ENABLES BETWEEN IC1 + IC2

Software NEWS & REVIEWS

ROBOT WARS (64K MSX Cassette)

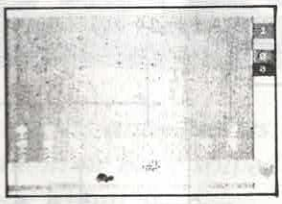
The androids have all gone crazy! They're trigger happy and you are in the firing line. You must zap then before they get you, but be careful, the corridors that you pass through are electrified, so no leaning on the wall for a quiet rest!



Based on the old Atari game, Berzerk. 1 player, several screens, bad music, average graphics, smooth movement.

Publisher: Aackosoft
Price: $14.95

THE HARE AND TORTOISE (SV318/328 Cassette)



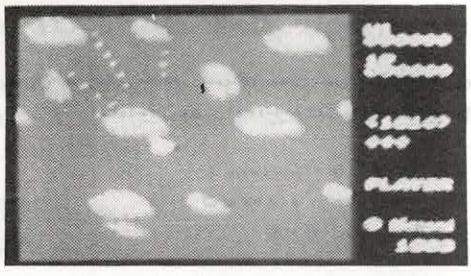
HARE & TORTOISE

Do you remember the famous race between the Hare and the Tortoise? Well, here it is again, only this time you are the Hare and you must count how many objects appear in rows across the screen. If you get it right, you hop on to the next column and so on to get to the prize (a big fat strawberry). Watch out for old Tortoise, though! He's not as slow as he looks.

1 player game , 1 screen, 9 levels - an innovative way to teach children to count.

Publisher: Spectravideo
Price: $10.00

TIME PILOT (MSX Cartridge)



Your fighter travels through four time zones and is attacked by various enemies. In the 1910 period you shoot down biplanes and airships and avoid enemy fire. Pick up the parachutes for extra points. The 1940 era brings monoplanes and bombers and in 1970 you are attacked by helicopters and homing missiles. Stage four, which we haven't reached, is the UFO era.

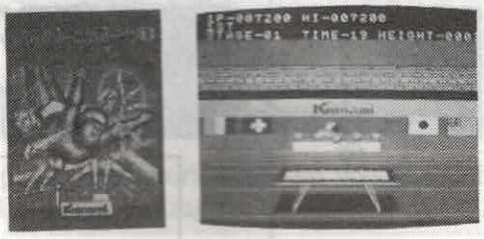
At first this game seemed dull although the scrolling and action were smooth. After a few attempts to reach higher levels, the challenge got me and I could not leave it alone - a pity these games are only on loan.

1 or 2 players, good graphics and music - addictive!

Publisher: Konami
Price: $49.95

Review copy supplied by LPG Software

HYPER SPORTS I (MSX Cartridge)



You are an Olympic athlete and you must perform to the best of your ability for your country. You must compete in a total of four events - diving, vaulting, trampolining and gymnastics on the high bar.

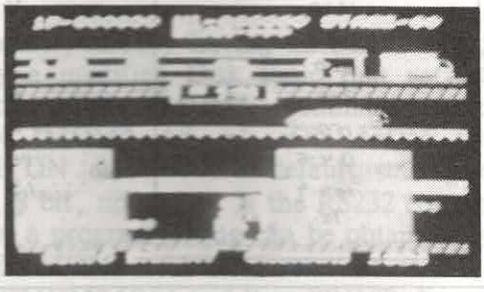
In each event you are given a qualifying score which you must match or beat before being allowed to compete in the next event. You are allowed three attempts at each event.

1 or 2 players, reasonable graphics and music - good but hard.

Publisher: Konami
Price: $49.95

Review copy supplied by LPG Software

COMIC BAKERY (MSX Cartridge)



Joe, the Baker, works 9 to 5 making bread for his bakery. The trouble is, he is plagued by the mischievous Raccoon Dogs which turn off his machines and conveyor belts so that they can steal the bread. Joe must zap the dogs with his hypnotic ray gun and turn the machines and belts back on again so that he can get at least four loaves baked for his regular, five o'clock customers in the bakery.

1 or 2 players, good graphics and music - excellent!

Publisher: Konami
Price: $49.95

Review copy supplied by LPG Software

BASIC TUTORIAL - Part 11

There was an error in the explanation of the TIME command last month. It was stated that the value of the memory location was incremented once every sixty seconds. It should have read once every one-sixtieth of a second or sixty times per second.

The ERASE command

This command is used to erase an array which was created using a DIM command and recover the memory space used by that array. It takes the form -

ERASE A$

where A$ is the name of a previously dimensioned array.

This command can be used when an array is not needed later in a program and the memory space is required for something else. It can also be useful if an array needs to be made larger or smaller. If you try to change the size of an array without first ERASEing it, you will get a 're-dimensioned array' error.

Try this -

```
10 PRINT FRE(0)
20 DIM A$(500)
30 PRINT FRE(0)
40 ERASE A$
50 PRINT FRE(0)
```

The DEF FN command

This is the DEFINE FUNCTION command. It can be used to define a variable as a formula or function.

Example

```
10 DEF FNA=B*2+10
20 INPUT "ENTER 1 - 10";B
30 PRINT FNA
40 GOTO 20
```

Anywhere in the program following the DEF FNA statement, FNA will be equal to the current value of B*2+10

SCREEN CONTROL

The CLS command

This is the CLEAR SCREEN command. In text mode (SCREEN 0) it performs exactly the same function as the CLS key on the keyboard and will clear everything from the screen and place the cursor at the top-left (home) position. In MSX graphics mode (SCREEN 1, 2 or 3) it will only clear the text and graphics from the screen - sprites will remain on screen. The command SCREEN 0 should be used to remove sprites from the screen. In SV Screen 1 or 2 CLS will clear text, graphics and sprites.

MSX Example 1

```
10 SCREEN 1
20 S$=STRING$(8,255) : SPRITE$(1)=S$
30 PRINT "SCREEN 1"
40 PUT SPRITE 1,(128,96),1,1
50 FOR T=1 TO 1000 : NEXT T
60 CLS
70 PRINT "CLEARED"
80 GOTO 80
```

MSX Example 2

```
10 SCREEN 2 : OPEN"GRP:"AS#1
20 S$=STRING$(8,255):SPRITE$(1)=S$
30 PRESET(0,0):PRINT#1,"SCREEN 2"
40 PUT SPRITE 1,(128,96),1,1
45 LINE(0,40)-(255,40)
50 FOR T=1 TO 1000 : NEXT T
60 CLS
70 PRESET(0,120):PRINT#1,"CLEARED"
80 GOTO 80
```

MSX Example 3

```
10 SCREEN 3 : OPEN"GRP:"AS#1
20 S$=STRING$(8,255):SPRITE$(1)=S$
30 PRESET(0,0):PRINT#1,"SCREEN 3"
40 PUT SPRITE 1,(128,96),1,1
45 LINE(0,40)-(255,40)
50 FOR T=1 TO 1000 : NEXT T
60 CLS
70 PRESET(0,120):PRINT#1,"CLEARED"
80 GOTO 80
```

SV Example 1

```
10 SCREEN 1
20 S$=STRING$(8,255):SPRITE$(1)=S$
30 LOCATE 0,0:PRINT"SCREEN 1"
40 PUT SPRITE 1,(128,96),1,1
45 LINE(0,40)-(255,40)
50 FOR T=1 TO 1000 : NEXT T
60 CLS
70 LOCATE 0,120:PRINT"CLEARED"
80 GOTO 80
```

SV Example 2

```
10 SCREEN 2
20 S$=STRING$(8,255):SPRITE$(1)=S$
30 LOCATE 0,0 : PRINT"SCREEN 2"
40 PUT SPRITE 1,(128,96),1,1
45 LINE(0,40)-(255,40)
50 FOR T=1 TO 1000 : NEXT T
60 CLS
70 LOCATE 0,120:PRINT"CLEARED"
80 GOTO 80
```

The WIDTH command

This command is used to set the WIDTH of the screen in text mode or MSX graphics screen 1 and when executed will clear the screen. It has no effect in MSX screen 2 or 3 and SV screen 1 or 2.

The valid range for MSX SCREEN 0 is 1 - 40 and SCREEN 1 is 1 - 32. The width is always centred. That is, WIDTH 20 in SCREEN 0 will have 10 columns either side of the centre line of the screen.

The valid range for SV SCREEN 0 is 39 - 40

MSX Example 1

```
10 SCREEN 0
20 WIDTH 40
30 PRINT "WIDTH"
40 GOTO 40
```

SV Example 1

```
10 SCREEN 0
20 WIDTH 40
30 PRINT "WIDTH"
40 GOTO 40
```

MSX Example 2

```
10 SCREEN 0
20 WIDTH 1
30 PRINT "WIDTH"
40 GOTO 40
```

SV Example 2

```
10 SCREEN 0
20 WIDTH 39
30 PRINT "WIDTH"
40 GOTOT40
```

MSX Example 3

```
10 SCREEN 1
20 WIDTH 32
30 PRINT "WIDTH"
40 S$=STRING$(8,255) : SPRITE$(1)=S$
50 PUTSPRITE 1,(128,96),1,1
60 GOTO 60
```

MSX Example 4

```
10 SCREEN 1
20 WIDTH 1
30 PRINT "WIDTH"
40 S$=STRING$(8,255) : SPRITE$(1)=S$
50 PUTSPRITE 1,(128,96),1,1
60 GOTO 60
```

The CSRLIN command

This is the CURSOR LINE command. It is used to determine the vertical position of the cursor on the screen and will return a number between 0 and 22.

Example 1

```
10 CLS
20 PRINT CSRLIN
```

Displays 0 because the CLS command places the cursor on line 0.

Example 2

```
10 CLS : LOCATE 0,20
20 PRINT CSRLIN
```

Displays 20 because the LOCATE command places the cursor on line 20.

The POS command

This is the cursor POSITION command. It takes the form -

POS (0)

The 0 is a dummy variable. It is used to determine the horizontal position of the cursor on the screen and will return a number between 0 and the value of the current screen width.

Example

```
10 CLS
20 PRINT POS(0)
```

Displays 0 because the CLS command places the cursor at the leftmost position of 0.

Example

```
10 CLS : LOCATE 20,0
20 PRINT POS(0)
```

Displays 20 because the LOCATE command places the cursor at position 20 of line 0.

PRINTER COMMANDS

The LPRINT command

This is the LINE PRINT command. The LINE in this case comes from the term LINE PRINTER which was the original name given to a computer printer. It performs in much the same way as the PRINT command and it is possible to write a program using the PRINT command and then when all is working properly, add an L to each PRINT command.

The LLIST command

This is the LINE LIST command. This is functionally the same as the LIST

command except that output is to the printer instead of the screen. The same parameters are used for LLIST as for LIST -

LLIST will print the full listing.
LLIST 100 will print line 100.
LLIST -100 will print all lines up to and including 100.
LLIST 100- will print all lines from and including 100.
LLIST 100-200 will print from line 100 to 200 inclusive.

PROGRAMMING TECHNIQUE

TEXT ON GRAPHIC SCREENS

Using text on the graphic screens is different for SV and MSX. To use text on the SV screens 1 and 2, it is only necessary to position the invisible graphics cursor, using LOCATE, to the screen position where the text is to be displayed using the PRINT command. For example -

```
100 SCREEN 1
110 LOCATE 128,96
120 PRINT "ABC"
130 GOTO 130
```

For MSX the procedure is a little more complex. First a graphics file buffer must be opened and all text written to it. The LOCATE command does not work in graphics mode and must be replaced by PRESET. For example -

```
100 SCREEN 2
110 OPEN "GRP:" AS #1
120 PRESET (128,96)
130 PRINT#1,"ABC"
140 GOTO 140
```

A further complication arises when we want to accept text input from the keyboard while the program is in graphics mode. In text mode, INPUT or INPUT$ could be used to accept the text. These commands will destroy a graphics screen because they force the computer back to text mode.

The alternative is to use INKEY$ for input. This is fine if you only require a one character response such as Y or N or a single digit. To accept a string of characters, such as a name, it is necessary to use a string input routine built around the INKEY$ command. For example -

MSX version

```
10 SCREEN 2,2
20 OPEN "GRP:" AS #1
30 LINE(10,10)-(245,181),5,BF
40 PRESET (31,100) : PRINT#1,"NAME"
50 X=72
```

```
60 I$=INKEY$ : IF I$="" THEN 60
70 PRESET (X,100) : X=X+8
80 IF I$<>CHR$(13) THEN PRINT#1,I$ : GOTO 60
90 GOTO 90
```

Line 20 opens the file buffer
Line 30 puts graphics on the screen
Line 40 prints the prompt
Line 60 waits for I$ to have a value
Line 70 locates the graphics cursor
Line 80 checks if the key is ENTER - if not, display the character and go back for the next

SV version

```
10 SCREEN 1,2
20 LINE(10,10)-(245,181),5,BF
30 LOCATE 31,100 : PRINT "NAME"
40 X=72
50 I$=INKEY$ : IF I$="" THEN 50
60 LOCATE X,100 : X=X+8
70 IF I$<>CHR$(13) THEN PRINT I$ : GOTO 50
80 GOTO 80
```

Line 20 puts graphics on the screen
Line 30 prints the prompt
Line 50 waits for I$ to have a value
Line 60 locates the graphics cursor
Line 70 checks if the key is ENTER - if not, display the character and go back for the next

CONVERTING MSX PROGRAMS TO SV AND BACK

The similarities between MSX Basic and Spectravideo Basic create a situation which is just begging for someone to write a machine code conversion program. Until that happens, we must do it manually.

CONVERTING MSX TO SV

BEFORE ATTEMPTING TO CONVERT ANY MSX LISTINGS TO SV (OR VICE VERSA) BE SURE THAT YOU HAVE HAD A HIGH SUCCESS RATE AT ENTERING PROGRAMS THAT WERE WRITTEN FOR YOUR COMPUTER WITHOUT MORE THAN A COUPLE OF TYPING ERRORS.

Make a careful study of the MSX listing. The first thing to look for are the SCREEN commands. The Spectravideo uses SCREEN 0 for text, SCREEN 1 and 2 for graphics and sprites. MSX uses SCREEN 0 for text (no problem there), SCREEN 1 for text and sprites, SCREEN 2 and 3 for graphics and sprites. If the program contains a SCREEN 1 command, then it will be almost impossible to convert it because the Spectravideo does not use

this screen mode -

MSX SCREEN	EQUIVALENT SV SCREEN
0	0
1	NONE
2	2
3	3

Another thing to watch for are graphics characters which are entered from the keyboard. The Spectravideo uses a LEFT GRAPH and RIGHT GRAPH key to produce 52 block graphics from the alphabetical keys on the keyboard. The MSX, on the other hand uses a GRAPH key and SHIFT/GRAPH to produce 87 block graphics and a CODE key and SHIFT/CODE to produce a range of non-English letters and symbols. There are a few similarities between the block graphics used by the two machines.

COMPARISON CHART OF SV AND MSX BLOCK GRAPHICS CHARACTERS

SV Character		MSX Character	SV Character		MSX Character
160 (LEFT+A)	= 20	(GRAPH+F)	183 (LEFT+X)	= 17	(GRAPH+B)
161 (LEFT+B)	= 223	(SHFT/GRP+I)	184 (LEFT+Y)	= 214	(SHFT/GRP+H)
162 (LEFT+C)	= 27	(GRAPH+N)	185 (LEFT+Z)	= 26	(GRAPH+V)
163 (LEFT+D)	= 19	(GRAPH+H)	187 (RIGHT+B)	= 3	(SHFT/GRP+,)
164 (LEFT+E)	= 25	(GRAPH+Y)	188 (RIGHT+C)	= 29	(GRAPH+/)
165 (LEFT+F)	= 222	(SHFT/GRP+K)	189 (RIGHT+D)	= 28	(GRAPH+X)
166 (LEFT+G)	= 193	(SHFT/GRP+D)	190 (RIGHT+E)	= 30	(GRAPH+\)
167 (LEFT+H)	= 221	(GRAPH+K)	191 (RIGHT+F)	= 9	(GRAPH+0)
173 (LEFT+N)	= 211	(SHFT/GRP+N)	192 (RIGHT+G)	= 199	(GRAPH+D)
176 (LEFT+Q)	= 24	(GRAPH+R)	198 (RIGHT+M)	= 5	(GRAPH+,)
177 (LEFT+R)	= 212	(SHFT/GRP+F)	199 (RIGHT+N)	= 4	(SHFT/GRP+;)
178 (LEFT+S)	= 21	(GRAPH+G)	201 (RIGHT+P)	= 219	(GRAPH+O)
179 (LEFT+T)	= 220	(GRAPH+U)	207 (RIGHT+V)	= 6	(GRAPH+;)
181 (LEFT+V)	= 213	(SHFT/GRP+V)	209 (RIGHT+X)	= 23	(GRAPH+-)
182 (LEFT+W)	= 18	(GRAPH+T)	211 (RIGHT+Z)	= 22	(SHFT/GRP+\)

If the program does not contain a SCREEN 1 command or any graphics characters from the keyboard, then it is probably quite safe to go ahead and type the listing in - provided you can be sure that it worked in the original MSX version.

If the program listing has been reproduced from a dot matrix printout, then there is every chance that it is an unadulterated listing of a program that runs. Be cautious of listings where there are not slashes through the zeroes in the line numbers. If they look like a capital letter O then you will have great difficulty in some cases determining whether it is an O or a zero in the body of the listing. Even worse are listings which have been produced on a typewriter which uses a lower case letter l for the number 1.

DISK BASIC

USING RANDOM ACCESS FILES

This is the real beauty of using disks. When we use a cassette based system, we can only use SEQUENTIAL access files. That is, to read a file it must be done sequentially, one record after another until the desired record is found. Sequential files can also be used with disk systems. If we want to change the information contained in a record, the whole file must be loaded into memory, the record details changed and the whole file written back to the tape or disk.

A random access file is made up of a number of individual records. For example a file containing names and addresses could be called NAMES, and each name and address within that file is called a record and each record has a number according to its position in the file. Random access allows you to read and write to a specific record within the file without having to look at all of the records that precede it in the file.

The OPEN command

When using the OPEN command with random access files, you do not need to specify the direction (input or output). It takes the form -

```
OPEN "filename" AS #1        (MSX format)
OPEN "d:filnam" AS #1        (SV format)
```

In MSX format, filename is the name of the file you wish to open (up to eight characters). In SV format, d: is the drive number (1: or 2:) and filnam is the name of the file (up to six characters) to be opened.

The FIELD command

With sequential files, the information is read from the file with something like INPUT#1,A$ and written to the file with something like PRINT#1,A$.

With random access, the information is taken from the disk, using the GET command and stored in a buffer or string variable. Information is written to the disk from this buffer using the PUT command.

The LSET command

This command is used to LEFT-SET a string variable within the buffer variable. It takes the form -

```
LSET X$=A$
```

This has the effect of placing A$ into X$ using spaces to pad out the trailing space. For example, if X$ is thirty characters long and A$ is twenty five characters long, then LSETting A$ into X$ will make X$ equal to A$ followed by five spaces. If A$ was longer than X$ then the extra characters to the right are lost.

The RSET command

This command is used to RIGHT-SET a string variable within the buffer variable. It takes the form -

```
RSET X$=A$
```

This places A$ into X$ using spaces to pad out the leading spaces. To use the same example as LSET, then RSETting A$ into X$ would make X$ equal to five spaces followed by A$. If A$ was longer than X$ then the extra characters to the right are lost.

The PUT command

This command is used to place a record, which has been entered into the fields of the buffer, on to disk. It takes the form -

```
PUT #N,X
```

where N is the buffer number and X is the number of the record on the disk.

The GET command

This command is used to place a record from the disk and place it into the fielded buffer. It takes the form -

```
GET #N,X
```

where N is the buffer number and X is the number of the record on the disk.

The CVI command

This is the CONVERT TO INTEGER command. Because all information used in random access files has to be stored in a string variable, then any numeric information will be retrieved in string form and therefore unusable in calculations. This command converts the string data to an integer variable. It takes the form -

```
N=CVI(A$)
```

where A$ is the field in the buffer which contains the numeric data and N is the variable which will be used in numeric calculations.

The CVS command

This is the CONVERT TO SINGLE-PRECISION command. It takes the form -

```
N=CVS(A$)
```

where A$ is the field in the buffer which contains the numeric data and N is the variable which will be used in numeric calculations.

The CVD command

This is the CONVERT TO DOUBLE-PRECISION command. It takes the form -

```
N=CVD(A$)
```

where A$ is the field in the buffer which contains the numeric data and N is the variable which will be used in numeric calculations.

The MKI$ command

This is the MAKE INTEGER command. Because the buffer for random access files is a string variable, all numeric data must be converted to a string before it can be stored in the buffer. It takes the form -

```
X$=MKI$(N)
```

where N is the number to be converted and X$ is the field in the buffer.

The MKS$ command

This is the MAKE SINGLE-PRECISION command. It takes the form -

```
X$=MKS$(N)
```

where N is the number to be converted and X$ is the field in the buffer.

The MKD$ command

This is the MAKE DOUBLE-PRECISION command. It takes the form -

```
X$=MKD$(N)
```

where N is the number to be converted and X$ is the field in the buffer.

EXAMPLE USE OF THESE COMMANDS

The following example makes use of the OPEN, FIELD, LSET, PUT, GET, MKI$ and CVI command. It requests input of eight names, addresses and phone numbers and saves these into the random access file called FILE. It then allows you to search each record by inputting the number of that record.

```
10 REM MAKE ROOM FOR STRINGS
20 CLEAR 1000
30 REM OPEN FILE
40 OPEN "1:FILE" AS #1 : REM MSX = 40 OPEN "FILE" AS #1
50 REM FIELD THE BUFFER
60 FIELD #1,8 AS A1$,30 AS A2$,30 AS A3$,30 AS A4$,30 AS A5$
70 REM LOOP FOR RECORD CREATION
80 FOR I=1 TO 8
90 CLS
100 PRINT "RECORD #";I
110 INPUT "NAME: ";NM$
120 INPUT "STREET: ";ST$
130 INPUT "SUBURB: ";SU$
140 INPUT "PHONE: ";PH$
150 REM PLACE IN BUFFER FIELDS
160 LSET A1$=MKI$(I)
170 LSET A2$=NM$
180 LSET A3$=ST$
190 LSET A4$=SU$
200 LSET A5$=PH$
210 REM PUT BUFFER TO DISK
220 PUT#1,I
230 NEXT I
240 REM SEARCH RECORDS BY NUMBER
250 CLS : PRINT "SEARCH ..."
260 INPUT "RECORD NUMBER 1-8";R
270 REM LOAD INTO BUFFER FROM DISK
280 GET#1,R
290 RN=CVI(A1$)
300 PRINT "RECORD #";RN
310 PRINT "NAME: ";A2$
320 PRINT "STREET: ";A3$
330 PRINT "SUBURB: ";A4$
340 PRINT "PHONE: ";A5$
350 PRINT
360 INPUT "ANOTHER RECORD (Y/N)";I$
370 IF I$="Y" OR I$="y" THEN 250
380 REM FINISHED WITH FILE SO
390 REM CLOSE BUFFER
400 CLOSE #1
```

ANSWERS to the questions in issue 10

1. (b) SCREEN 0 is a text screen

2. POKEing 4000 will have no effect as it is in the ROM area and the contents of ROM cannot be changed. The ROM extends from memory location 0 to 32767.

3. 24 because this was the value of B before it was SWAPped with A.

4. It will display -
   ABCDE
   12345
   20 PRINT "12345"

5. Yes - because A is a numeric variable and ABC requires a string variable.

QUESTIONS

1. What is wrong with using this on a Spectravideo?

   ```
   WIDTH 20
   ```

2. Does the CLS command clear sprites from the MSX screen?

3. Could this MSX program be easily converted to SV format?

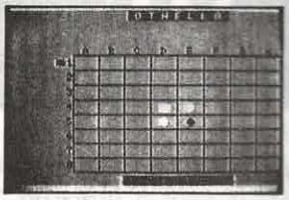
   ```
   10 SCREEN 1
   20 PRINT "SCREEN 1"
   ```

4. Is the following used to open a random MSX or SV file?

   ```
   OPEN "NEWFILE" AS #1
   ```

5. Which command must be used before you can enlarge or reduce the size of an array?

OTHELLO (SV318/328 Cassette)



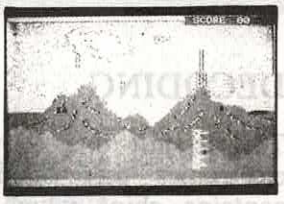
OTHELLO SD238T

This is a tough, 'let's get down to business' type of game, where you pit your wits against the computer, striving to win the computerised version of the age-old board game - Othello. It's a game of skill, employing tactics to gain an advantage. So, put on your thinking caps!

1 player, average graphics.

Publisher: Spectravideo
Price: $12.00

RESCUE (SV318/328 Cassette)



You have been assigned to rescue 3 stranded men who infiltrated enemy territory to obtain secret plans. You must fly your two-seater helicopter over rough terrain and firing guns to collect these men one at a time. Sharpen your reflexes and off you go!

1 player, fairly good graphics, good sound effects and concept. 1 screen, 3 levels.

Publisher: Spectravideo
Price: $15.00

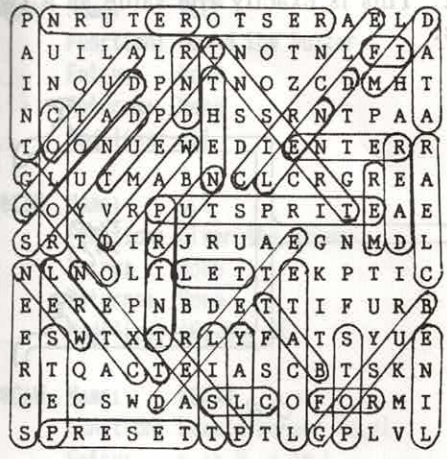
COMPUTER KIDS

Hi kids! This month we have the answer to the last word search and a little program which will help the little kids with their counting. Don't forget to make the changes to line 50 if you have an MSX computer.

Bye for now, Wendy.

```
10 CLS : PRINT"GUESS THE NUMBER OF SQUARES"
20 PRINT"type in the number and press ENTER"
40 PRINT : N=INT(RND(1)*9)+1
50 A$=CHR$(201)+CHR$(32)
60 FOR I=1 TO N : PRINTA$; : NEXT I
70 PRINT : PRINT
80 INPUT"How many squares";A
90 IF A<>N THEN 110
100 PRINT"Correct!" : GOTO 40
110 PRINT"There are "N" squares"
120 GOTO 40
```

COMPUWORDS



MSX owners change line 50 to read -

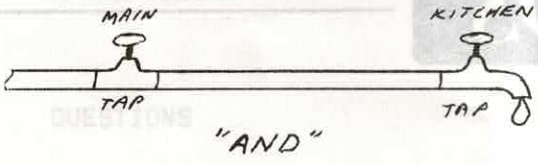
```
50 A$=CHR$(219)+CHR$(32)
```

UNDERSTANDING MACHINE CODE

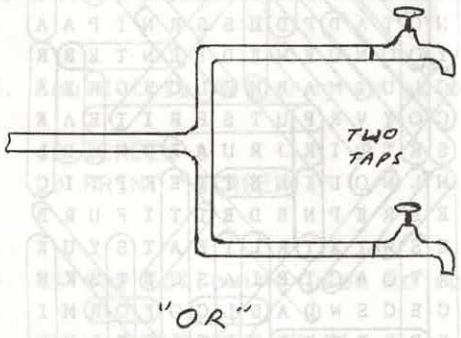
DECODING

This is just one of the ways in which digital electronics uses binary maths. The computer contains chips which control the direction of signals coming from the CPU, the keyboard, etc. Some of these are called AND gates and OR gates.

A two input AND gate has two input signals and one output signal and requires BOTH inputs to have a signal (or one) before it will pass a signal (or one). Let us use the example of taps in a house. If the main tap at the meter is ON, AND the tap in the kitchen is ON, then there will be water coming into the sink (assuming you have paid your bill).



If we ignore the main tap at the meter and had two cold water taps (A and B) in the kitchen, then turning on A OR B OR both, will put water in the sink. This is exactly the same as a two input OR gate.



THE Z80 INSTRUCTION SET

The Z80 CPU has a set of instructions which are used to write machine code programs. This can be compared with the commands used by a Basic interpreter. The Basic commands can be single-word commands (eg CLS) or multiple commands which require you to provide some further information (eg ON N GOTO 500).

HINTS & TIPS

Ron Nicholls of St Kilda sends in these tips which are for MSX but you might like to try them with your SV318/328.

I have no idea if this discovery is original, but the Draw commands will accept negative numbers. ie U-10 will draw down, R-10 will draw left. This is almost like the rotate instruction, but easier to keep track of and more interesting.

The listing at the end of the program is a short fast way to read the joystick.

```
10 GOTO30
20 FORD=0TO500:NEXT:CLS:RETURN
30 SCREEN2
40 DRAW"BM125,100S4"
50 FORT=0TO8STEP.1:DRAW"U=A;R=B;"
70 A=INT(SIN(T)*5):B=INT(COS(T)*5)
80 NEXT
100 GOSUB20
110 A=10:DRAW"BM5,80S4"
130 FORT=0TO20:DRAW"U=A;R10"
140 A=A*-1:NEXT
160 GOSUB20
170 A=10:B=10:DRAW"BM130,80"
190 FORT=0TO20:DRAW"U=A;R=B;"
200 A=A*-1-4:B=B+-1:NEXT
220 GOSUB20
230 A=10:B=10:DRAW"BM20,100"
250 FORT=0TO20:DRAW"D=A;H=B;R10"
260 A=A*-1:B=B*-1:NEXT
280 GOSUB20
290 A=10:C=4:DRAW"BM20,90"
310 FORT=0TO20:DRAW"S=C;E=A;F=A;R10"
320 A=A*-1:C=C+3:NEXT
330 GOSUB20
340 SCREEN0
350 LIST 360-430
360 ' ROUTINE TO READ STICK
370 ' SET UP ARRAY
380 DIMQ(8,1)
390 FORT=0TO8:READQ(T,0),Q(T,1):NEXT
400 DATA 0,0,0,-1,1,-1,1,0,1,1,0,1,-1,1,
-1,0,-1,-1
410 'READ STICK
420 X=X+Q(STICK(0),0)
430 Y=Y+Q(STICK(0),1)
```

HACKERS CORNER

ACCESSING SV ROM ROUTINES

This routine makes use of the 'Locate GRP cursor' routine at &H48E9 and the 'Draw Line' routine at &H4AF9. It draws individual lines to form a solid block. The equivalent Basic program is also given for comparison of speed. Try them both.

```
10 CLEAR 200,&HD000
20 FOR I=&HD000 TO &HD00F
30 READ A$ : POKE I,VAL("&H"+A$)
40 NEXT I
50 Y=10
60 SCREEN 1
70 DEFUSR=&HD000
75 POKE&HD004,Y
80 J=USR(0)
85 IF Y<180 THEN Y=Y+1 : GOTO 75
90 GOTO 90
100 DATA 01,0A,00 :'LD BC,10 (X loc)
110 DATA 11,0A,00 :'LD DE,10 (Y loc)
120 DATA CD,E9,48 :'CALL 48E9 posn csr
130 DATA 21,80,00 :'LD HL,128 line len
140 DATA CD,F9,4A :'CALL 4AF9 set line
150 DATA C9       :'RET
```

and in Basic...

```
10 SCREEN 1
20 FOR Y=10 TO 180
30 LINE(10,Y)-(138,Y)
40 NEXT Y
```

THE SV ROM DECODED

Spectravideo SV-318/328 ROM version 1.1

```
7227  RSET
7228  LSET
72CD  FIELD
7312  MKI$
7315  MKS$
7318  MKD$
732B  CVI
732E  CVS
7331  CVD
7375  CLOSE
73AD  LFILES
73B2  FILES
73CA  EXTWR
73F1  EXTRD
7406  INPUT
7484  LOC
749A  LOF
74B0  EOF
74C6  FPOS
7624  BSAVE
762A  Get start address for above
7635  Get end address for above
7643  Get execution address
765D  Open the file
7662  Write start address
7668  Write end address
766E  Write execution address
7672  Write memory to device
7684  BLOAD
76A1  Get loading address
76B2  Open file
7788  I/O lookup table
77CC  KYBD device table
7817  CRT device table
```

MSX ROM CALLS

```
00BDH  Name:       CKCNTC
       Function:   Same as ISCNTC
       Entry:      None
       Returns:    None
       Modifies:   None

00C0H  Name:       BEEP
       Function:   Sound the buzzer
       Entry:      None
       Returns:    None
       Modifies:   All

00C3H  Name:       CLS
       Function:   Clear the screen
       Entry:      None
       Returns:    None
       Modifies:   AF, BC, DE

00C6H  Name:       POSIT
       Function:   Locate cursor (x,y)
       Entry:      x in H, y in L
       Returns:    None
       Modifies:   AF

00C9H  Name:       FNKSB
       Function:   Check function key display.
                   If active then display keys
       Entry:      FNKFLG
       Returns:    None
       Modifies:   All
```

BOOKS • BOOKS

THE MSX RED BOOK
Reviewed by Ian Davidson

I have just received my much awaited copy of the MSX Red Book. I purchased it from the UK by mail order. The book is designed for the serious machine code programmer. It describes the MSX hardware and software at a level of detail only previously available to the computer manufacturer. It is well presented, well indexed and easy to read despite being a reference book. It assumes an understanding of machine language and thankfully does not waste valuable space duplicating information already available in Z80 reference books.

There are seven sections covering all aspects of the MSX. The first three are short, and are perhaps there only for completeness. That is no loss as these chapters - on the Programmable Peripheral Interface (PPI), the Video Display Processor (VDP) and Programmable Sound Generator (PSG) are well covered by other literature.

The Red Book comes into it's own in the next chapters describing the ROM BIOS, the interpreter and the memory organisation.

The ROM BIOS (Basic Input/Output System) is covered in the next 63 pages. A functional description of every separate routine in the ROM, between 0000H and 268BH is given, and is easily readable. A good example is one of the paragraphs on the Video RAM quoted here -

Address...07CDH
Name.....WRTVRM
Entry.....A=Data byte, HL=VRAM address
Exit.....None
Modifies..EI

Standard routine to write a single byte to the VDP VRAM. The VRAM address is first set up via the SETWRT standard routine and then the data byte written to the VDP data port. Note that the two seemingly spurious EX(SP),HL instructions in this routine, and several others, are required to meet the VDP's timing constraints.

The section on the ROM BASIC Interpreter is equally comprehensive, taking a further 119 pages. It is presented in a similar style and is best demonstrated by quotes, firstly from the section introduction -

> One point to note when examining the interpreter in detail is that it contains a lot of trick code. The writers seem particularly fond of jumping into the middle of instructions to provide multiple entry points to a routine. As an example take the instruction:
>
> 3E D1 Normal: LD A,0D1H
>
> When encountered in the usual way this will of course load the accumulator with the value D1H. However if it is entered at "normal + 1" then it will be executed as a POP DE instruction. The interpreter has many similar obscure sections.

Another shorter description of one of the routines -

> Address...4A1DH
>
> This is the "LPRINT" statement handler. PRTFLG is set to 01H, to direct output to the printer, and control transfers to the "PRINT" handler (4A29H).

The final section of the book, 25 pages, gives some examples of assembled programs which use the previously described ROM features.

The MSX Red Book is a valuable book for anyone working on the more serious side of programming - machine language, or who just wants to know more about his computer.

Published by KUMA Computers Ltd., UK. Review copy purchased by mail order from Richard Reeves, UK.

Written any programs? Send them in to COMPUTER FORUM for inclusion in our Program Listings. If we decide to publish, you will receive payment based on size, originality and scope of interest. Please send programs in cassette form or disk and clearly marked which format they are for. If you wish them to be returned, enclose sufficient postage for safe return. Mark your name and address on everything.

PROGRAM LISTINGS

Each issue of COMPUTER FORUM will contain program listings for you to type in. When entering these programs it is a good idea to save off to cassette or disk every 10 or 20 lines and then continue. Then if the power goes off or another major disaster occurs, you will not have to type in the whole lot again.

Another check is to verify each save using the CSAVE? command. To use this command, CSAVE your program, then rewind the tape and type CLOAD? and press enter. This command will then compare the program on the tape with the one in memory, to verify that it was saved correctly.

All of the listings are printed in 40 column format. If you enter WIDTH40 before typing them in, you will be able to line up the characters in the listing with the text on screen.

Take care that you do not mistake the letter l for 1 or an O for a zero. Where graphics characters are used in the program, they will be shown as @1@. This symbol will then be repeated at the end of the listing with instructions concerning the graphics character used.

All of the listings have been tested on MSX and SV318/328 computers and do work. Should you have any problems, we suggest that you thoroughly check your work as errors can creep in very easily. Should you not wish to type in these listings, send a blank tape together with $2 to COMPUTER FORUM and we will post a copy to you. Do not forget to tell us which computer you have.

MINI-RACING
by Alex Wilson

A very compact racing program which, although it will not challenge Le Mans, it is very addictive.

MSX ONLY - Because this program uses MSX Screen 1, it is not easily converted to Spectravideo format.

```
1 DATA040444E4440404040404040424440420
20202024 2A242020202020202020
2 DATA8484858585858585858585858585848400
00F01010101010 1F01010101FF0000
3 DATA00360836002 22A3E0002023E001E241E21
21212121212121212121212121212121
4 DATA8484848484844242212110100808040400
003E2222223E000000808040402020
5 DATA0000000000018244242241901020204042 1
21212121214242848408081010202 0
6 DATA0404040404040202010110381000000020
20202020201010080884844242212 1
7 DATA0404040404040808101021214242848420
20202020204040809824241800000 0
8 DATA8080808080804040202010100808040420
20242A2420202020202024282428 20
```

```
9 DATA8484848484848282818180808080808000
0000181800000183C3C988040402020
10 DATA0404040404040808101020204040402
0202026242424262020202620202020
11 DATA8080808080808080808080808080802
020242A2420202020202026262020202
12 IFVPEEK(14336)<>4THENSCREEN1,3:FORI=0
TO10:READA$:S$="":FORJ=1TO64STEP2:B$=MID
$(A$,J,2):S$=S$+CHR$(VAL("&H"+B$)):NEXT
J:SPRITE$(I)=S$:NEXTI:SPRITE$(11)=CHR$(1
28)
13 SCREEN1:WIDTH26:KEYOFF:COLOR4,15,1:VP
OKE8196,4:VPOKE8219,164:ONSPRITEGOSUB21:
SPRITEON:X=144:FORI=1TO300:NEXT
14 FORI=7TO14:LOCATE12,I:PRINTSTRING$(6,
219):NEXTI:LOCATE4,0:PRINT"************
":LOCATE4,1:PRINT"*MINI-RACING*":LOCATE4
,2:PRINT"************":LOCATE2,20:PRINT
"C 1986 Alex Wilson"
15 LOCATE0,7:PRINT"STAGE  1":LOCATE0,9:P
RINT"SCORE  0":LOCATE0,11:PRINT"HIGH":PR
INT"SCORE ";HS
16 IFNOT(STRIG(0)+STRIG(1))THEN16
17 FORS=0TO3:LOCATE6,7:PRINTS+1:N=INT(RN
D(-TIME)*5)+1:ONNGOSUB24,25,26,27,28:REA
DB,C:FORD=0TO12:A=B:B=C:READC:E=86:LOCAT
E6,9:PRINTD*32+S*384:FORE=86TO117STEP2+S
/2
18 PUTSPRITE3,(X,100),1,11:F=STICK(0)+ST
ICK(1):IFF=3THENX=X+1ELSEIFF=7THENX=X-1
19 SPRITEOFF:PUTSPRITE0,(128,E),4,A:PUTS
PRITE1,(128,E-32),4,B:PUTSPRITE2,(128,E-
64),4,C:SPRITEON:NEXTE,D,S
20 LOCATE0,14:PRINT"WELL DONE":SOUND7,25
4:PLAY"v15o4l64e":FORI=1TO100:NEXT:PLAY"
gaaaaaa":HS=1569:LOCATE6,9:PRINTHS:FORI=
1TO2000:NEXT:GOTO13
21 IFTIME<10THENGOSUB23:LOCATE0,14:PRINT
"<CRASH>":H=INT(D*32+E-86+S*384):LOCATE6
,9:PRINTH:FORI=1TO1000:NEXT:IFH>HSTHENHS
=H:GOTO13ELSEGOTO13
22 TIME=0:RETURN
23 SOUND6,31:SOUND7,247:SOUND8,16:SOUND1
2,20:SOUND13,0:RETURN
24 RESTORE29:RETURN
25 RESTORE30:RETURN
26 RESTORE31:RETURN
27 RESTORE32:RETURN
28 RESTORE33:RETURN
29 DATA0,0,3,1,6,0,4,2,5,7,10,8,6,0,0
30 DATA0,0,3,6,4,5,7,8,6,7,8,6,0,0,0
31 DATA0,0,7,9,4,2,2,5,7,8,6,3,6,0,0
32 DATA0,0,4,5,3,6,3,1,1,9,4,5,0,0,0
33 DATA0,0,7,10,10,10,8,6,7,8,6,4,5,0,0
```

DUCK SHOOT

This listing is in MSX format. For SV318/328 owners, see note at the end of the listing.

```
10 REM ducks
20 DEFINT A-Z:WIDTH 40:SCREEN 2,2
30 OPEN "GRP:" AS #1:LINE(15,15)-(240,17
6),15,BF
40 COLOR 1:PRESET(96,16):PRINT#1,"DUCK S
HOOT"
50 PRESET(24,32):PRINT#1,"Shoot the duck
s as they sit      on the water."
60 PRESET(24,64):PRINT#1,"Use the joysti
ck or cursor       keys to move your guns
ight       around and press the fire
  button to score."
70 PRESET(24,104):PRINT#1,"Make sure the
 duck is fully      in your sights."
80 PRESET(64,160):PRINT#1,"ENTER LEVEL 1
-5"
90 I$=INKEY$:IF I$<"1"ORI$>"5" THEN 90
100 L=VAL(I$)
110 COLOR 15
120 ON INTERVAL=500/L GOSUB 320
130 ON STRIG GOSUB 350:STRIG(0)ON:STRIG(
1)ON
140 FOR I=1 TO 32:READ A
150 S$=S$+CHR$(A):SPRITE$(1)=S$
160 NEXTI
170 SPRITE$(2)=CHR$(255)+STRING$(14,128)
+CHR$(255)+CHR$(255)+STRING$(14,1)+CHR$(
255)
180 CLS:LINE(0,184)-(255,191),1,BF
190 FOR I=1 TO 50:L1=RND(1)*200:L2=RND(1
)*180:LINE(L1,L2)-(L1+16,L2):NEXT I
200 FOR I=1 TO 12:L1=RND(1)*200:L2=RND(1
)*180:LINE(L1,L2)-(L1,L2+16),1:CIRCLE(L1
,L2-4),4,10,,,5/1:NEXT I
210 PUT SPRITE 1,(90,96),1,1
220 PUT SPRITE 2,(128,96),1,2
230 INTERVAL ON
240 FOR I=0 TO 8:READ X1(I),Y1(I):NEXT
250 PUT SPRITE 1,(D1,D2),C,1
260 D=STICK(0)+STICK(1)
270 G1=G1+X1(D):G2=G2+Y1(D)
280 IF G1<0 THEN G1=0 ELSE IF G1>224 THE
N G1=224
290 IF G2<0 THEN G2=0 ELSE IF G2>160 THE
N G2=160
300 PUT SPRITE 2,(G1,G2),1,2
310 GOTO 250
320 READ D1,D2:IF D1=999 THEN RESTORE 43
0:P=P+1:IF P=4 THEN 500 ELSE 320
330 C=INT(RND(1)*15)+1:IF C=4 THEN C=5
340 DK=1:RETURN
350 IF DK=0 THEN RETURN
360 IF G1=D1 AND G2=D2 THEN BEEP:SC=SC+2
0:LINE(0,184)-(255,191),1,BF:PRESET(60,1
84):PRINT#1,"SCORE = ";SC:DK=0
370 RETURN
380 DATA 0,0,28,244,60,24,31,31
390 DATA 63,63,63,31,0,0,0,0
400 DATA 0,0,0,0,0,255,255,255,254,252
410 DATA 248,240,0,0,0,0
420 DATA 0,0,0,-8,8,-8,8,0,8,8,0,8,-8,-8
,-8,0,-8,-8
430 DATA 160,160,32,32,224,64,64,32,128,
96,192,160,0,0,96,96
440 DATA 224,128,96,160,64,64,192,0,128,
64,192,32,128,128,96,0
450 DATA 32,0,32,160,0,64,128,96,160,128
,32,128,64,96,128,160
460 DATA 64,0,32,64,64,160,192,128,160,3
2,96,32,0,96,32,96
470 DATA 128,0,96,64,64,128,128,32,160,9
6,224,160,224,32,224,0
480 DATA 160,0,192,64,160,64,192,96,0,12
8,0,160,0,32,224,96
490 DATA 999,0
500 SCREEN 0
510 LOCATE 10,10:PRINT"Your score was ";
SC
520 LOCATE 10,15:PRINT"Play again (y/n)"
530 I$=INKEY$:IF I$="" THEN 530
540 IF I$="Y" OR I$="y" THEN RUN
550 END
```

NOTE: SV318/328 owners, change lines 20,30,40,50,60,70,80, and 360 to read -

```
20 DEFINT A-Z:WIDTH 40:SCREEN 1,2
30 LINE(15,15)-(250,176),15,BF
40 COLOR 1:LOCATE 96,16:PRINT"DUCK SHOOT
"
50 LOCATE 24,32:PRINT"Shoot the ducks as
 they sit on the          water."
60 LOCATE 24,64:PRINT"Use the joystick o
r cursor keys to          move your gunsi
ght around and press          the fire but
ton to score."
70 LOCATE 24,104:PRINT"Make sure the duc
k is fully in your          sights."
80 LOCATE 64,160:PRINT"ENTER LEVEL 1-5"
360 IF G1=D1 AND G2=D2 THEN BEEP:SC=SC+2
0:LINE(0,184)-(255,191),1,BF:LOCATE 60,1
84:PRINT"SCORE = ";SC:DK=0
```

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